WORKSHEET 1.12

Unit 1 Free-Response Practice: Scoring Guide

Unit 1: Limits and Continuity · 18 points

Question 1 · 9 points

(a) For $x < 2$, $\dfrac{x^2 - x - 2}{x - 2} = \dfrac{(x-2)(x+1)}{x-2} = x + 1$, so $\lim\limits_{x \to 2^-} f(x) = 3$. **[1 point: left-hand limit with supporting work]**

$\lim\limits_{x \to 2^+} f(x) = \sqrt{2 + 7} = 3$. **[1 point: right-hand limit]**

(b) $k = 3$. f is continuous at $x = 2$ when $\lim\limits_{x \to 2^-} f(x) = \lim\limits_{x \to 2^+} f(x) = f(2)$, and both one-sided limits equal 3. **[1 point: $k = 3$; 1 point: justification using the limits]**

(c) A removable discontinuity, because $\lim\limits_{x \to 2} f(x) = 3$ exists but $f(2) = 5 \neq 3$. **[1 point: type with reason]**

(d) For large x, $f(x) = \sqrt{x + 7}$, so $\lim\limits_{x \to \infty} \dfrac{\sqrt{x+7}}{\sqrt{x}} = \lim\limits_{x \to \infty} \sqrt{1 + \dfrac{7}{x}} = 1$. **[1 point: uses the correct piece; 1 point: answer]**

(e) Yes. With $k = 3$, f is continuous on $[2, 9]$. $f(2) = 3$ and $f(9) = \sqrt{16} = 4$, so $f(2) < 3.5 < f(9)$. By the Intermediate Value Theorem there is a c in $(2, 9)$ with $f(c) = 3.5$. **[1 point: continuity and $f(2) < 3.5 < f(9)$; 1 point: conclusion naming the theorem]**

Question 2 · 9 points

(a) $\dfrac{W(6) - W(2)}{6 - 2} = \dfrac{46 - 52}{4} = -1.5$ megalitres per day. **[1 point: difference quotient and value]**

From $t = 2$ to $t = 6$ days, the amount of water in the reservoir decreased at an average rate of 1.5 megalitres per day. **[1 point: interpretation with units and interval]**

(b) W is continuous on $[0, 10]$. 50 lies between $W(0) = 40$ and $W(2) = 52$, between $W(2) = 52$ and $W(6) = 46$, and between $W(6) = 46$ and $W(10) = 61$. By the Intermediate Value Theorem, $W(t) = 50$ at least once in each of $(0, 2)$, $(2, 6)$ and $(6, 10)$: at least 3 times. **[1 point: continuity stated; 1 point: 50 between the values on the three intervals; 1 point: answer 3]**

(c) $\lim\limits_{t \to \infty} \left(60 - 20e^{-0.3t}\right) = 60$, since $e^{-0.3t} \to 0$. **[1 point: limit]**

In the long run the model predicts that the amount of water in the reservoir approaches 60 megalitres. **[1 point: interpretation]**

(d) With $t = 0.01$: $\dfrac{M(0.01) - 40}{0.01} = 5.991$. With $t = -0.01$: 6.009. With $t = \pm 0.001$: 5.999 and 6.001. The limit is approximately 6.000. **[1 point: values on both sides of 0; 1 point: estimate]**

Using your score

Mark each question out of 9. As a rough guide for a single unit: 14 or more out of 18 is strong; 10 to 13 is on track; below 10, rework the lessons named beside the points you missed (limits: 1.5; continuity: 1.8; discontinuity types: 1.7; limits at infinity: 1.9; Intermediate Value Theorem: 1.10; rates and interpretation: 1.1; tables: 1.3).

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